# Input Ontology (IO)

## Score: 3/5 — Moderate gaps  |  Confidence: medium

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Definition: Whether the benchmark's test case categories cover the query types expected in deployment.

## Justification

MILU's taxonomy of 8 domains and 41 subjects spans the breadth needed for a multi-subject teacher-evaluation deployment, and the format is exclusively single-correct-answer MCQ, which is a direct structural match for the deployment's binary positive/negative marking requirement. However, the taxonomy is derived from competitive exam content (UPSC/SSC/state PSC) rather than school-board syllabi, and the dataset sample shows a heavy skew toward Engineering & Tech (23% of Hindi validation items) with under-representation of Civics/Political Science and Panchayati Raj content that is central to UP/MP/Rajasthan/Bihar state-board curricula. Per-subject statistics for Hindi specifically are not publicly available, making it impossible to verify whether the categories adequately cover state-board curricular framing.

## Strengths

- Exclusive single-answer MCQ format directly matches the deployment's binary scoring requirement [Q22, Q33, DATASET-D1]
- All 8 declared domains appear in the Hindi validation sample, providing nominal multi-domain breadth [DATASET-D1, D2, D4, D9, D20]

| ID | Evidence |
|---|---|
| D1 | जब एक डीसी सीरीज मोटर बिना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English |
| D2 | निम्नलिखित में से कौन सा खनिज उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question |
| D4 | राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? 44वें संविधान संशोधन अधिनियम द्वारा — Constitutional law MCQ — Law & Governance domain |
| D9 | निम्नलिखित में से कौन प्रत्यक्ष कर का उदाहरण है? संपत्ति कर — Economics/taxation MCQ — pan-India competitive exam register |
| D20 | मई 2022 में इराक के सुलेमानिया में आयोजित तीरंदाजी एशिया कप 2022 स्टेज 2 अभियान में भारत ने कितने स्वर्ण पदक जीते? आठ — Sports GK — India-centric, recent current affairs |

- Indic LLM sub-taxonomy is treated separately, indicating awareness of system heterogeneity [Q60]

## Checklist

### Checklist item definitions:

| ID | Assessment Question |
|---|---|
| IO-1 | Identify test case categories required for regional deployment |
| IO-2 | Check if taxonomy omits regionally relevant categories |
| IO-3 | Check if taxonomy includes irrelevant categories |
| IO-4 | Document category gaps harming content validity |

### Checklist responses:

IO-1: Required categories for the deployment include Hindi Literature (canonical board authors), History (regional medieval and freedom movement), Civics/Political Science with Panchayati Raj and state administration, Geography of North India, Science/Maths in Hindi, Economics, and General Knowledge. MILU declares all 8 domains and 41 subjects [Q42] which nominally cover this breadth, with subject-level statistics in supplementary Table 9 [Q98].

IO-2: MILU's taxonomy is derived from competitive exam content [Q23, Q24, Q27] rather than school/university board syllabi. State-specific civics (Panchayati Raj, state legislatures) is not visible as a dedicated category in the sample [DATASET-D4 is the only Law & Governance item, and concerns the 44th constitutional amendment, not state-level civics]. The single dedicated Hindi-board literature register (close reading of Tulsidas/Premchand/Kabir) appears absent from the sample [DATASET-D5 — the only Literature item is about Wordsworth].

IO-3: Engineering & Tech dominates the Hindi validation sample (6 of 26 items) including DC motors, transformers, rectifiers, and FORTRAN [DATASET-D1, D13, D14, D23], which are characteristic of polytechnic/engineering entrance content rather than secondary or degree-college teacher domains. This represents construct-irrelevant variance for a general Hindi-medium teacher deployment.

| ID | Evidence |
|---|---|
| **D1** | जब एक डीसी सीरीज मोटर बिना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English |
| **D13** | हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: पल्सेटिंग डीसी — Electrical engineering MCQ — translated from English |
| **D14** | स्थिर धारा ट्रांसफार्मर _______ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration |
| **D23** | फॉर्ट्रान 77 के फिक्स्ड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉर्मेट लेबल — Computer science MCQ — translated, technical vocabulary |

IO-4: Identified gaps: (a) school-literature framing for Hindi canonical authors absent; (b) state-specific civics/Panchayati Raj under-represented; (c) over-representation of engineering content; (d) per-language per-subject distributions not published [NOT FOUND per regional YAML state_wise_exam_source_distribution_for_hindi].

| ID | Evidence |
|---|---|
| **WEB-9** | HuggingFace dataset is gated, limiting independent verification of per-subject Hindi distributions |
| **WEB-12** | GitHub subject categories include 'Literature and Linguistics' and 'Arts and Culture' but no item-level framing detail published |

## Information Gaps

- Per-language per-subject item counts for Hindi not published in accessible form (Table 9 in gated PDF)
- Whether the test split's domain distribution matches the sampled validation distribution
- Whether MILU includes any items framed in school-literature register vs. competitive-exam GK

## Requires Expert Verification

- Whether the 41-subject taxonomy maps cleanly onto UP/MP/Rajasthan/Bihar board syllabi at the subject level
- Whether the missing categories (state civics, Hindi literary close reading) are material to teacher use cases or marginal

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## Remediation

**Gap 1: Heavy skew toward Engineering & Tech (23% of Hindi sample) and under-representation of Civics/Political Science, Panchayati Raj, and state-board literature framings**

Recommendation: Use MILU domain-level results as a coarse signal only; build a deployment-specific subject inventory that mirrors UP/MP/Rajasthan/Bihar board syllabi, and weight or filter MILU subjects to align with actual teacher use cases